Hall Effect

Yukun (Eric) Cao
Lab Partner: Adriano Rasmussen
Northeastern University
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We investigated the electrical properties of a doped silicon wafer using four-wire resistance and Hall effect measurements. A Hall sample with six indium contacts was prepared to allow current application along the sample length and voltage measurements across two regions. Resistivity values of $7.820 \pm 0.205\ \Omega \cdot \text{cm}$ and $7.535 \pm 0.197\ \Omega \cdot \text{cm}$ were determined using four-wire techniques. Hall voltage measurements across the sample width yielded a carrier density of $(2.544 \pm 0.071) \times 10^{15}\ \text{cm}^{-3}$ and mobilities of 313.74 ± 12.01 and $325.62 \pm 12.47\ \text{cm}^2/\text{V} \cdot \text{s}$, confirming the sample as p-type. The observed asymmetry in I–V behavior was consistent with non-ohmic conduction arising from Schottky barrier effects, contact resistance, and Seebeck-induced thermoelectric voltages.

INTRODUCTION

The Hall effect is a fundamental tool for characterizing semiconductor materials. When an electric current flows through a sample placed in a perpendicular magnetic field, the Lorentz force deflects moving charge carriers—electrons (negative) or holes (positive)—to one side, generating a Hall voltage transverse to both current and field directions. This voltage depends on the sign and density of charge carriers, enabling determination of carrier type, concentration, and mobility from measurable quantities [1, 4, 5].

Real semiconductor samples, however, often exhibit deviations from ideal behavior. At the metal–semiconductor interface, charge redistribution leads to the formation of a Schottky barrier, which impedes current flow. Under moderate to strong electric fields, this barrier can be partially overcome via *enhanced carrier injection* (the electric field helps more carriers cross the barrier) or *barrier suppression* (the barrier becomes effectively thinner or lower in height) [2, 3]. These mechanisms produce nonlinear I–V behavior and cause resistance to vary with applied voltage, particularly under reverse bias. Asymmetries in the I–V curve can thus be attributed to differences in contact quality—such as solder pancaking, contamination, or contact geometry—that result in unequal barrier heights at opposite terminals. In this experiment, indium was used for its ability to form low-resistance ohmic-like contacts to silicon, though surface oxidation and incomplete wetting can still introduce *contact-limited conduction*, where current flow is constrained by interface resistance rather than bulk resistivity.

Thermoelectric effects also influence measurements. When a temperature gradient exists across a semiconductor, the Seebeck effect generates a voltage due to the diffusion of hot carriers from the heated region to the cooler region [6, 7]. This can lead to baseline offsets in Hall voltage measurements that do not pass through the origin, especially in the presence of thermally unbalanced leads or contact junctions.

This experiment combines the Hall effect with four-wire resistivity measurements to determine the carrier concentration and mobility in a doped silicon wafer. By comparing inner voltage contact pairs and analyzing asymmetries in I–V and R–V behavior, we investigate the roles of Schottky barriers, carrier injection, and contact geometry in shaping semiconductor transport behavior.

APPARATUS

- Doped silicon wafer
- Pre-wired circuit board
- Carbide scriber
- Two soldering irons
- Indium solder
- Lead-tin solder
- Rubber cement
- Fine wire
- Electromagnet, GMW Magnet Systems, Model 3470
- DC power supply, 2 current sources, GW Instek GPS-3030DD (±0.03% + 4 digits)
- Two digital multimeters, GW Instek GDM-8145 (±0.2% + 2 digits)
- Gaussmeter, Magnetic Technology Inc. WT103 (±1%, 0.01 mT resolution)
- Dial caliper (resolution: 0.01 mm)
- MATLAB software

PROCEDURES AND RESULTS

A. Each Group Makes a Hall Sample from a Silicon Wafer

To investigate charge transport properties in a doped silicon wafer, each group assembled a Hall sample and confirmed proper electrical contacts were made. A pre-cut silicon wafer segment was selected with six indium contacts already deposited. Using a dial caliper, the sample thickness was measured to be $d = 0.39 \pm 0.01$ mm, and its width was found to be $w = 8.87 \pm 0.01$ mm. The length between contact pairs 2–3 and 5–6 was $L_{23} = 7.57 \pm 0.01$ mm and $L_{56} = 7.63 \pm 0.01$ mm, respectively.

The sample was mounted to a pre-wired circuit board using a small amount of rubber cement. Care was taken to ensure that the silicon surface was not contaminated. Fine wires were then soldered from the board to the indium contacts on the silicon wafer using separate soldering irons and two types of solder (lead-tin and indium respectively) to prevent cross-contamination. Wire labeling and contact mapping were confirmed and recorded.

After assembly, all 6 contact pair resistance combinations were measured using a digital multimeter to ensure electrical continuity and proper soldering contact. All measured values were well below 1 MΩ, confirming reliable electrical contact. These results are summarized in Table I.

TABLE I. Measured resistance between contact pairs (in kΩ).

From → To	2	3	4	5	6
1	0.977	0.612	1.185	0.644	0.417
2		5.320	5.625	5.360	5.215
3			0.813	0.602	0.613
4				1.123	1.222
5					0.645

To characterize the sample's electrical behavior, a current was applied between contacts 1 and 4 using a DC power supply, and both the applied voltage and resulting current were measured using two digital multimeters. The measurements were repeated in both polarities to identify any asymmetry or nonlinearity. Current-voltage (I-V) data were collected up to a power limit of $P = IV < 0.1$ W to minimize Joule heating.

The resulting I-V curve is plotted in Fig. 1. The response is approximately linear on the forward-bias (positive voltage) side, but somewhat curved under reverse bias. There is also a noticeable asymmetry between positive and negative voltages which indicates that the sample does not exhibit fully ohmic behavior. While the positive branch suggests near-ohmic conduction at low fields, the steeper slope and greater current suggest enhanced carrier injection in this direction. In contrast, the reverse-bias region shows a suppressed current and clear nonlinearity, implying that additional barriers, such as contact resistance or depletion regions, may inhibit carrier transport in that direction.

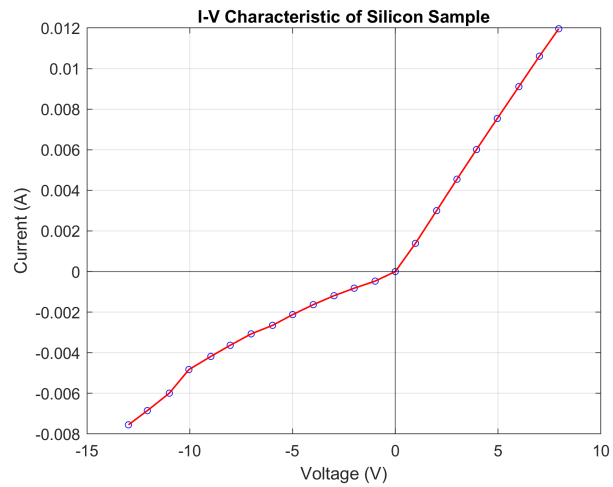

FIG. 1. Current-voltage (I–V) characteristic of the Hall sample. Forward bias shows near-linear behavior; reverse bias is curved and less conductive.

Because the sample is a doped semiconductor rather than a metal, deviations from linearity are expected due to voltage-dependent mobility, carrier depletion at contacts, and asymmetries at the metal-semiconductor junctions. The current–voltage asymmetry is consistent with potential Schottky-like behavior at the contacts, where one direction facilitates carrier injection while the other suppresses it. Furthermore, because power was limited to $P < 0.1$ W, a higher voltage could be applied in reverse bias than forward, amplifying the nonlinearity observed on the negative end.

The corresponding resistance curve is plotted in Fig. 2, where $R = V/I$ was computed from the same data. In forward bias, the resistance remains relatively constant, further supporting near-ohmic conduction under small positive voltages. However, in reverse bias, R decreases significantly with increasing magnitude of voltage. This trend reflects enhanced carrier injection or barrier suppression at higher fields, characteristic of semiconductor conduction mechanisms.

B. 4-wire Method for Measuring Resistivity of a Doped Silicon Wafer

To determine the resistivity of the two sides of the doped silicon wafer, a current was applied across the outer contacts (1 and 4) using a DC power supply. The voltage drops across the two inner contact pairs (2–3 and 5–6) were measured using two digital multimeters to be 1.559 ± 0.005 V and 1.514 ± 0.005 V, respectively. This constitutes a 4-wire measurement, in which separate voltage-sensing leads minimize the influence of contact

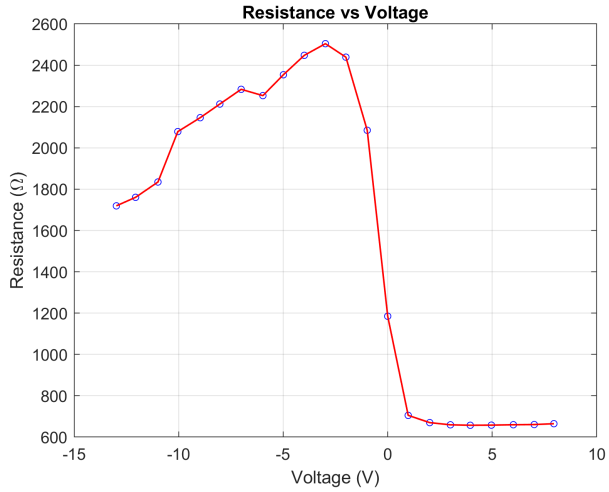

FIG. 2. Resistance versus applied voltage for the Hall sample. Reverse bias shows clear field dependence, while forward bias is flatter.

resistance and lead wire resistance on the measurement. The resistance was calculated from Ohm's law, $R = V/I$, using the applied current value $I = 9.11 \pm 0.04$ mA.

Using the measured distances $L_{23} = 7.57 \pm 0.01$ mm and $L_{56} = 7.63 \pm 0.01$ mm, and sample cross-sectional area $A = w \cdot d = (8.87 \pm 0.01 \text{ mm}) \cdot (0.39 \pm 0.01 \text{ mm}) = (3.459 \pm 0.089) \times 10^{-6} \text{ m}^2$, the resistivity was computed as:

$$\rho = R \cdot \frac{A}{L}. \tag{1}$$

For the 2–3 pair, the measured resistance was $R_{23} = 171.13 \pm 0.87\ \Omega$, yielding a resistivity of:

$$\rho_{23} = 0.07820 \pm 0.00205\ \Omega \cdot \text{m} = 7.82 \pm 0.21\ \Omega \cdot \text{cm}.$$

For the 5–6 pair, $R_{56} = 166.19 \pm 0.85\ \Omega$ gave:

$$\rho_{56} = 0.07535 \pm 0.00197\ \Omega \cdot \text{m} = 7.54 \pm 0.20\ \Omega \cdot \text{cm}.$$

To contrast this with the 2-wire method, the power supply was disconnected and an ohmmeter was used to measure resistance directly across the same contact pairs. The 2-wire resistance readings were $R_{23}^{(2w)} = 1191\ \Omega$ and $R_{56}^{(2w)} = 615\ \Omega$, significantly larger than the 4-wire values. This discrepancy arises because in the 2-wire method, the same leads are used to both source current and measure voltage, meaning the total resistance includes contributions from the probe wires, connectors, and especially the contact resistance at the indium-silicon interface.

The 4-wire technique eliminates these effects by isolating the voltage measurement from the current path, allowing the voltage drop to reflect only the intrinsic sample resistance.

C. Thermoelectric Determination of Carrier Type

To determine the carrier type of the silicon sample, a voltmeter was connected across the outer contacts, with contact 1 connected to the positive terminal. A hot soldering iron tip was brought momentarily near each contact to induce a local thermal gradient. When contact 1 was heated, the voltmeter reading decreased (became more negative), while heating contact 4 caused the reading to increase (became more positive).

This behavior is consistent with diffusion of majority carriers in a **p-type** semiconductor. Heating a region increases the kinetic energy of carriers—holes, in this case—causing them to diffuse from hot to cold regions. When contact 1 is heated, holes diffuse away from it toward contact 4, leaving behind a region of lower potential at the hot end (1), resulting in a negative voltage reading from 1 to 4. Conversely, heating contact 4 causes holes to diffuse toward the cooler region at contact 1, raising the potential at the hot end (4) and yielding a positive voltage from 1 to 4.

This thermally driven diffusion confirms that the majority charge carriers are holes, indicating that the silicon wafer is p-type.

D. Hall Effect, Carrier Concentration and Mobility

The Hall effect provides a means to determine both the sign and density of charge carriers in a semiconductor. When a current I_S flows through a sample in the presence of a perpendicular magnetic field B, the Lorentz force deflects the moving charge carriers, causing them to accumulate on one side. This buildup of charge creates an electric field across the sample width—perpendicular to both I_S and B—resulting in a measurable Hall voltage V_H:

$$V_H = \frac{BI_S}{ned}, \tag{2}$$

where n is the carrier density, e is the elementary charge, and d is the thickness of the sample.

To determine the Hall voltage dependence on magnetic field, the magnetic field B was first calibrated using a Gaussmeter by measuring B for a range of magnet currents I_M from approximately 0.5 A to 3 A. A linear fit was applied to the data using MATLAB, yielding a calibration factor of $\alpha = 0.15831 \pm 0.00012$ T/A. This was used to convert subsequent magnet current values into magnetic field strength. The calibration results are shown in Fig. 3.

$$B(I_M) = \alpha I_M \tag{3}$$

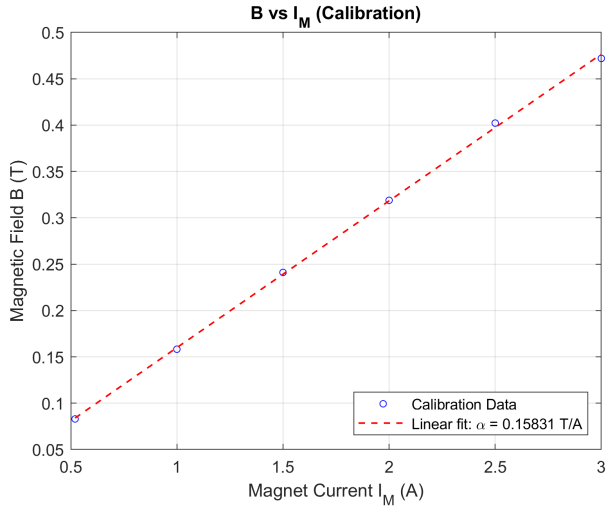

FIG. 3. Calibration of magnetic field B as a function of magnet current I_M.

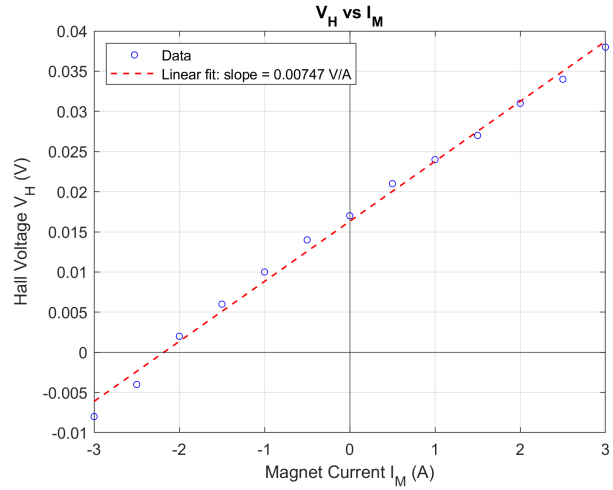

FIG. 4. Measured Hall voltage V_H as a function of magnet current I_M.

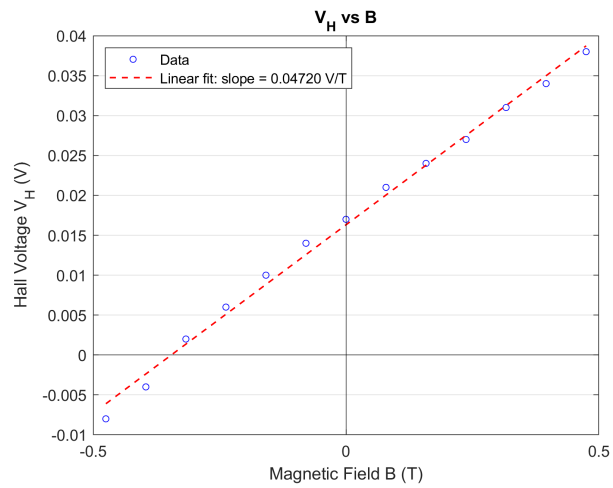

FIG. 5. Measured Hall voltage V_H versus magnetic field B. A linear fit was applied to determine the slope.

Next, a current $I_S = 7.51 \pm 0.03$ mA was applied across the outer contacts (1 and 4), keeping $P = IV < 0.1$ W. To select the most suitable Hall voltage measurement pair, voltages were measured across both contact pairs spanning the narrow width of the sample: contacts 2–6 and 3–5. The measured values were $V_{2-6} = 0.032 \pm 0.005$ V and $V_{3-5} = 0.016 \pm 0.005$ V. The 3–5 pair was chosen for subsequent Hall measurements due to its lower signal magnitude and improved signal-to-noise performance.

To measure the Hall effect, the sample was placed equidistant between the two magnetic pole faces. A constant sample current $I_S = 7.50 \pm 0.03$ mA was applied through the outer contacts, and the Hall voltage V_H was measured across contacts 3 and 5 for a range of magnet currents I_M. Measurements were taken in both field directions by reversing the magnet leads.

The resulting Hall voltage as a function of magnet current is shown in Fig. 4. This data was converted to Hall voltage as a function of magnetic field strength B using the calibration factor $\alpha = 0.15831$ T/A.

The $V_H(B)$ plot, shown in Fig. 5, displays a strong linear trend. However, the linear fit does not pass through the origin. This offset is commonly observed in Hall measurements and can result from background voltages, contact asymmetries, or thermoelectric effects within the wiring.

From the linear fit, the slope was determined to be $dV_H/dB = 0.04720 \pm 0.00047$ V/T. Using this value, the sample thickness $d = 0.39 \pm 0.01$ mm, and the sample current, the carrier density was computed from:

$$n = \frac{I_S}{e\, d\, (dV_H/dB)} \tag{4}$$

The resulting carrier density was found to be $(2.544 \pm 0.071) \times 10^{15}$ cm^{-3}.

To identify the carrier type, we examined the direction of the Hall voltage relative to the applied magnetic field and current. When facing the physical setup, current was defined as flowing out of the page when going positively through the silicon and the magnetic field from left to right when using positive current. The measured Hall voltage (potential difference) then increases measuring from contact 3 to 5, indicating a net Lorentz force in the positive vertical direction. As shown in Fig. 6, the diagram is rotated for visual clarity, with the magnetic field depicted as pointing out of the page rather than horizontally. Despite this, the relative vector directions are preserved. Note the voltmeter provides a positive reading when the positive lead is connected to a higher potential than the negative lead. Applying the right-hand rule to $\vec{F} = q(\vec{v} \times \vec{B})$, we find that the observed force direction is consistent with positively charged carriers,

confirming the sample is **p-type**.

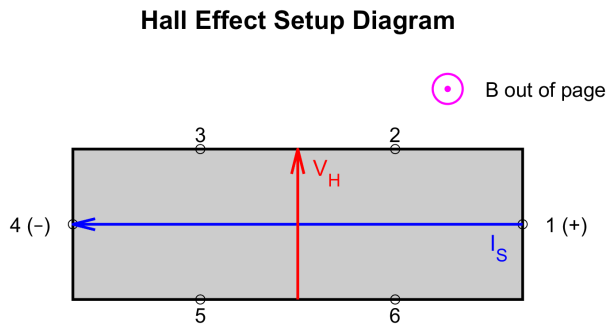

FIG. 6. Diagram of Hall measurement setup showing current, magnetic field, and Hall voltage directions.

Finally, carrier mobility was computed from the measured carrier density and the resistivity values obtained in Section B using

$$\mu = \frac{1}{ne\rho} \tag{5}$$

This yielded mobility values of $\mu_{23} = 313.74 \pm 12.01$ cm^2/Vs and $\mu_{56} = 325.62 \pm 12.47$ cm^2/Vs, consistent with expected values for moderately doped p-type silicon as further discussed in the summary.

SUMMARY

In this experiment, we characterized the properties of a doped silicon wafer using four-wire resistance measurements and the Hall effect. A Hall sample was soldered and verified for electrical contact integrity before measuring voltage and current behavior under controlled power limits. The I–V data showed slight nonlinearity and polarity-dependent asymmetry, consistent with contact-limited conduction and the influence of Schottky barriers. Resistance was calculated as a function of voltage and confirmed to vary with applied voltage, indicating enhanced carrier injection and barrier suppression in the reverse-bias region.

Resistivity was calculated independently for two voltage probe pairs using four-wire measurements, yielding values of $\rho_{23} = 7.820 \pm 0.205$ Ω·cm and $\rho_{56} = 7.535 \pm 0.197$ Ω·cm. The Hall effect was used to determine the type, density, and mobility of charge carriers. From the slope of the $V_H(B)$ curve, the carrier density was calculated as $n = (2.544 \pm 0.071) \times 10^{15}$ cm^{-3}, and the sample was confirmed to be p-type via the right-hand rule.

Using the measured resistivities and carrier density, The resulting mobilities were $\mu_{23} = 313.74 \pm 12.01$ cm^2/Vs and $\mu_{56} = 325.62 \pm 12.47$ cm^2/Vs. These values fall short of the expected range for moderately doped p-type silicon [1**?**].

TABLE II. Measured transport properties of the Hall sample.

Contact Pair	ρ (Ω·cm)	μ (cm^2/Vs)	Accepted μ_h [1]
2–3	7.820 ± 0.205	313.74 ± 12.01	450–500
5–6	7.535 ± 0.197	325.62 ± 12.47	450–500

Carrier density: $n = (2.544 \pm 0.071) \times 10^{15}$ cm^{-3} (from dV_H/dB slope)

Mobility in semiconductors is known to decrease as doping concentration increases, due to increased impurity scattering. As more dopant atoms are introduced into the lattice, they act as scattering centers, disrupting the path of mobile carriers and reducing their average drift velocity under an applied electric field. This relationship is particularly relevant in the low-to-moderate doping regime where lattice scattering and impurity scattering are the dominant mobility-limiting mechanisms.

In our experiment, the carrier density n was effectively constant across both 2–3 and 5–6 voltage contact pairs. The observed variation in mobility must then result from differences in measured resistivity ρ rather than differences in doping concentration.

These differences in ρ show us the largest source of error is likely form contact quality, solder adhesion, and geometric uncertainties between the two voltage contact pairs rather than material property variations.

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